Conditioning as Group Action: Exact Compositional Conditioning at FiLM Cost

Richard Bao
richardbao419@gmail.com

Abstract

Networks are conditioned by scaling features (FiLM) or by generating weights (hypernetworks). Both learn what each condition does. Neither learns what two conditions do together unless training data shows them together. We condition by rotating features, with rotation angles a linear function of the condition. Rotations add, so combined conditions compose automatically, and the network handles combinations it never saw. **CGA** (Conditioning as Group Action) formalizes this: the condition enters the operator linearly in an abelian Lie algebra, $T(c) = P\exp(\mathrm{A}(Wc))P^{\top}$, so $T(c_1{+}c_2) = T(c_1)T(c_2)$ holds for any weights. A diagonal algebra yields a compositional FiLM variant; the rotation algebra with scalar position is RoPE. Under a pre-registered protocol (margins, tests, and splits committed before every run; conditioning cost within 1.2× FiLM), CGA posts the lowest error on never-trained condition combinations across five transformation suites, with complete seed separation (p=9.1×10^{-5}, N=10) and zero-shot extrapolation to combinations longer than any combination seen in training. Swapping its linear map for an MLP raises error by up to 4.3×10^{4}×; the most expressive baselines generalize worst. Three registered gates returned negatives that bound the method: a fit penalty on rich scenes, a content-conditioning failure (rotations move information and cannot add it), multi-step rollouts, where every conditioner compounds error equally unless a latent-consistency loss is added, and — the sharpest bound — a frozen representation, where the same task CGA wins by 8× when the representation is co-trained is lost by 1.6× when it is not. The negatives specify the successor. **Complex FiLM**, magnitude for content and phase for composition, beats FiLM by 36% on unseen combinations and matches it on content (0.97×) under two further pre-registered gates, and on a 2D Navier-Stokes surrogate conditioned on physical parameters it passes every criterion at once: 64.6% below the best unstructured arm, every seed beating every seed, while fitting *better* in distribution (0.938×) at 0.84× FiLM's cost. The same structure yields a guidance knob needing no second forward pass, which distorts 2–5× less than classifier-free guidance on a deterministic target, though not on a multimodal one. One operator, both roles, FiLM cost.

1 Introduction

Nearly every conditional model must inject something into its computation: a class, an action, a timestep, a text embedding. Call this conditioning. Diffusion models modulate every block on the timestep and class via adaLN [2]; visual reasoning, speech, and RL systems modulate features on questions, speakers, and goals [1]; world models condition latent dynamics on actions. Almost all of this conditioning is carried by three mechanisms: concatenation, feature-wise affine modulation (FiLM), or a hypernetwork that emits weights from the condition.

Two of these bracket an expressivity axis; we set concatenation aside, and carry it as a baseline arm. FiLM's diagonal transform is cheap and stable but cannot mix features. Hypernetworks can express any linear operator but pay quadratic cost and, as we show, memorize the training conditions: their error on unseen combinations of familiar conditions is worse than FiLM's, despite 43× the conditioning compute. The middle of this axis, structured operator families between diagonal and dense, is well explored for weight fine-tuning [6–9] and largely unexplored in FiLM's role: per-sample, input-conditioned modulation of activations.

Conditioning plays two roles, and the distinction runs through this paper. A condition can name a *change* to apply, such as a pose offset or an edit, or it can name the *content* to produce, such as a class or a caption. FiLM serves both. We ask which structure, at FiLM-level cost, buys compositional generalization: correct behavior on combinations of conditions never seen together in training. Our answer imports a mechanism from positional encoding. RoPE [4] and its generalization GRAPE [5] encode position n as a group action $G(n) = \exp(n\,\omega L)$; because position enters linearly in the Lie algebra, $G(n{+}m) = G(n)G(m)$ holds exactly. We transplant this construction from scalar positions in attention to arbitrary condition vectors in activation modulation.

Contributions. **(1) Framework.** CGA treats conditioning as a group representation of the condition space; FiLM, RoPE, and GRAPE drop out as special cases, and a two-line proposition makes condition composition exact for any weights (§3). **(2) Protocol.** A pre-registered, budget-fair evaluation: margins, statistical tests, and single-read splits committed to version control before every run, with a shared counter enforcing parameter and FLOP parity; we report every verdict, including negatives against our own predictions (§4). **(3) Where it works, and where it stops.** CGA has the lowest error on unseen condition combinations in all five suites where the condition is a transformation, by up to five orders of magnitude, and in every image suite the more expressive the conditioner, the worse it generalizes. Three of the eight suites returned pre-registered negatives, and each names a setting where the composition law of Proposition 1 buys nothing: 3D Shapes, which CGA wins on unseen combinations but loses on training fit; content conditioning; and multi-step rollouts (§5). **(4) Successor and guidance.** The negatives specify Complex FiLM, validated under two further gates as a FiLM replacement covering both conditioning roles, whose phase channel also yields guidance as an exact operator power: 2–5× less distortion than CFG with no unconditional pass (§5).

2 Related work

FiLM and conditional normalization. FiLM [1] and its normalized variants (conditional BN/LN, adaLN in DiT [2]) apply per-feature affine transforms generated from a condition. All are diagonal. We keep FiLM's role and enrich only the operator.

Structured operators in parameter-efficient fine-tuning (PEFT). Low-rank, orthogonal, and butterfly operator families are well studied as weight fine-tuning: LoRA [6], OFT [7], BOFT [8], HRA [9], Group-and-Shuffle [10]; He et al. [11] give a unified view. These learn one fixed transform per task. Our setting differs on every axis that matters: per-sample, input-conditioned, activation-space, and evaluated for compositional generalization. Hypernetworks [3] generate weights from conditions and inherit the memorization behavior we quantify.

Position encodings as group actions. RoPE [4] rotates query-key pairs by angles linear in position; GRAPE [5] generalizes this to group representations with exact relative laws. The mechanism is theirs; our contribution is the transplant to arbitrary condition vectors in FiLM's role, with a learned basis, and the budget-fair compositional evidence.

Compositional generation. Montero et al. [12] showed reconstruction-trained generative models fail on unseen factor combinations. Score and energy composition [23] composes concepts across separate conditional passes; we compose conditions inside the operator, in one pass. Symmetry-based representation learning [13, 14] argues latents should carry group actions; we supply the conditioning-side mechanism.

Nearest prior work. We searched deliberately for work that would invalidate this paper. The closest hits: Feature-space rotations with exact composition exist for geometric transformations [17]; abelian exponential-map operators for disentanglement [22]; attribute operators for recognition [26]; multilinear attribute mixing for unseen combinations [27]; per-sample amplitude-and-phase activation modulation in periodic implicit networks [20]; and composition by complex phase addition in knowledge-graph embeddings [21]. Rotation-structured latent world models with latent-prediction losses originate with Quessard et al. [15], Caselles-Dupré et al. [16], and the homomorphism autoencoder [18], whose composition property is emergent rather than imposed; latent consistency objectives are standard in model-based RL [19]. Guidance has been moved into the conditioning path by distilling a CFG teacher [25], and CFG's high-scale distortion is established [24]. We claim none of these components. We claim the combination: a group-structured operator in FiLM's role at budget parity, composition imposed exactly by a linear-in-algebra map, magnitude split from phase as content and composition channels, guidance as an exact power, and verdicts from pre-registered gates.

3 The CGA framework

Let $x \in \mathbb{R}^d$ be features and $c \in \mathbb{R}^k$ a condition. FiLM computes $y = \gamma(c) \odot x + \beta(c)$; hypernetworks emit $W(c)$. All are instances of

$$y = T(c)\, x + \beta(c), \tag{1}$$

and they differ only in where the per-sample operator $T(c)$ comes from.

Definition 1 (Compositional conditioner)**.** *$T : \mathbb{R}^k \to \mathrm{GL}(d)$ is* compositional *if it is a homomorphism from $(\mathbb{R}^k, +)$: $T(c_1{+}c_2) = T(c_1)\,T(c_2)$ and $T(0) = I$.*

Proposition 1 (Exact composition by construction)**.** *Let $\mathfrak{a}$ be an abelian subalgebra of $\mathfrak{gl}(d)$, $\mathrm{A} : \mathbb{R}^m \to \mathfrak{a}$ linear, $W \in \mathbb{R}^{m\times k}$, and P orthogonal. Then $T(c) = P\exp(\mathrm{A}(Wc))\,P^\top$ is a compositional conditioner for every W, P. If $\mathfrak{a}$ is skew-symmetric, $T(c)$ is orthogonal.*

Proof. $\exp(X)\exp(Y) = \exp(X{+}Y)$ for commuting X, Y; $\mathrm{A}\circ W$ is linear; conjugation preserves products; $\exp(0) = I$. ◻

Composition is usually a behavior the network must learn from data. Here it is a property of the parametrization. It holds at initialization, during training, and far outside the training distribution. Training only decides which directions the condition rotates and how fast.

Special cases. Choices of $(\mathfrak{a}, c, P)$ recover deployed mechanisms:

$\mathfrak{a}$	condition	P	recovered mechanism
$\mathrm{diag}(\mathbb{R}^d)$	c	I	*exponential FiLM*: $y = e^{Wc} \odot x$ (compositional)
$\mathfrak{so}(2)^{d/2}$	$c = n$ (scalar position)	I	RoPE [4]
$\mathfrak{so}(2)^{d/2}$	$c = n$, learned planes	learned	multiplicative GRAPE [5]
$\mathfrak{so}(2)^{d/2}$	general $c \in \mathbb{R}^k$	structured	**this work** (FiLM's role)
$\mathrm{diag} \times \mathfrak{so}(2)^{d/2}$	general c	canonical	**Complex FiLM** (§5)

Standard FiLM reaches the same diagonal operators through an MLP head $\gamma(c)$. The operators match; only the composition law is missing, and §5 shows that its absence is what breaks FiLM on unseen combinations.

Cost. With $\mathfrak{a} = \mathfrak{so}(2)^{d/2}$, exp is closed-form planar rotation: $3d$ FLOPs, no matrix exponential. The basis P is the whole budget question. A dense learned P costs $4d^2$ FLOPs per sample, which is 1.52× FiLM's entire conditioning path at d=128. Two fixes work. A structured orthogonal P (five block-diagonal Cayley layers with fixed shuffles, following Group-and-Shuffle) brings the total to 1.11× FiLM, with 4,800 parameters against the 8,128 dimensions of $\mathrm{SO}(128)$; that is enough, because P only needs the planes the condition acts on. Inside a transformer, no explicit P is needed: the adjacent dense projections can absorb P, exactly as RoPE relies on the query and key projections rather than on a basis of its own. Linearity also gives the operator a strength dial for free: $T(c)^w = T(wc)$, so raising the operator to a power is the same as scaling the condition, and §5 uses this in place of CFG's extrapolation between a conditional and an unconditional prediction. The operator equals the identity at initialization and at c=0, and every singular value is 1. Unit tests pin all three properties. Complex FiLM extends the algebra to $\mathrm{diag}\times\mathfrak{so}(2)^{d/2}$: each feature pair is multiplied by $m\,e^{i\theta}$, with magnitude from FiLM's expressive head (the content channel) and phase linear in the condition (the composition channel), at 0.84× FiLM's cost in our harnesses.

4 Pre-registered, budget-fair protocol

Vocabulary. An *arm* is one conditioning method, trained under the shared recipe. A *suite* is one experiment: a fixed backbone, dataset, and training loop that all of its arms share. A *gate* is that suite's pass-or-fail rule, written down before the run as a conjunction of *criteria*: typically a margin over the strongest baseline on unseen combinations, a no-regression criterion on the training distribution, and a budget criterion. A suite passes only if all of its criteria pass, and we report the outcome either way.

Design. Every suite is pre-registered: success margins, statistical tests (N=10 seeds, one-sided Mann–Whitney $p \le 0.01$, Cliff's $|\delta| \ge 0.474$), and data splits were fixed and committed to version control before each run. Each test split is read once, after all selection on a validation split. All arms share one backbone and one training recipe. A shared counter enforces the budget: our conditioning path stays within 1.20× FiLM's FLOPs and 1.05× the smallest hypernetwork baseline's parameters; baselines may be larger, and reach 48× our parameter count. Deviations are dated amendments in the committed specifications, plus one accounting erratum that our own audit caught and that downgraded a favorable verdict.

Table 1: Error on never-trained condition combinations: mean (seed sd) over 10 seeds. CGA is lowest in every column, with complete seed separation against the strongest hypernetwork baseline (p=9.1×10^{-5}, δ=−1.0). OOD means out-of-distribution: a condition combination never seen in training. Every column except 3D Shapes passed its full pre-registered gate; 3D Shapes failed on a separate training-fit criterion (text). The last column reruns the dSprites setup with Complex FiLM. **Columns are not comparable to one another**: the two synthetic suites report operator MSE, the rest report per-pixel MSE.

Conditioner	Aligned	Hidden basis	dSprites	+shape	3D Shapes	Complex FiLM
FiLM	1.3 (0.1)×10^{-2}	1.6 (0.04)×10^{-2}	3.0 (0.2)×10^{-3}	5.9 (0.2)×10^{-3}	4.5 (1.0)×10^{-3}	3.1 (0.2)×10^{-3}
Concat-MLP	1.5 (0.3)×10^{-2}	1.1 (0.4)×10^{-2}	2.9 (0.1)×10^{-3}	6.3 (0.2)×10^{-3}	5.4 (5.2)×10^{-3}	–
Cond-LN	2.4 (0.1)×10^{-2}	2.7 (0.05)×10^{-2}	3.2 (0.2)×10^{-3}	6.1 (0.3)×10^{-3}	8.6 (13.1)×10^{-3}	–
Hypernetwork	1.9 (0.3)×10^{-3}	3.4 (0.9)×10^{-3}	3.8 (0.3)×10^{-3}	7.0 (0.4)×10^{-3}	9.3 (2.3)×10^{-3}	3.8 (0.2)×10^{-3}
Dyn. low-rank	5.7 (0.5)×10^{-3}	6.1 (0.4)×10^{-3}	5.1 (0.9)×10^{-3}	8.1 (0.4)×10^{-3}	2.0 (0.4)×10^{-2}	–
MLP-head (abl.)	–	1.1 (0.2)×10^{-3}	2.7 (0.2)×10^{-3}	5.0 (0.2)×10^{-3}	3.3 (0.8)×10^{-3}	–
Complex FiLM (linear mag.)	–	–	–	–	–	2.0 (0.1)×10^{-3}
Complex FiLM	–	–	–	–	–	2.0 (0.2)×10^{-3}
CGA (ours)	7.4 (2.0)×10^{-4}	3.3 (2.2)×10^{-8}	1.8 (0.1)×10^{-3}	4.0 (0.1)×10^{-3}	1.2 (0.2)×10^{-3}	1.8 (0.1)×10^{-3}

Threats to validity, and the design answer. The benchmarks are self-constructed, so we built them to remove judgment from scoring: dSprites and 3D Shapes are deterministic factor-to-image maps, so every held-out condition has a unique ground-truth image and pixel error is an objective compositional metric, with no perceptual score and no human rating. Pre-registration removes outcome-contingent analysis; the verdicts below include three negatives, one of which falsified our own headline prediction. Every number in this paper regenerates from the committed logs with one command per suite; re-running a suite from scratch costs one training run per arm per seed.

Arms. FiLM; conditional LayerNorm; Concat-MLP; a dense hypernetwork $I{+}\Delta W(c)$; a dynamic low-rank map $I{+}A(c)B(c)^{\top}$; the Lie operator; Complex FiLM (both variants); and an MLP-head ablation of our own method (same basis, same operator family, more compute, no linearity).

The experiments. **Two synthetic suites.** Learn $y = M(c)x$, where c picks a combination of 8 rotation primitives. The second hides the primitives behind a random basis, so the operator has to find them, and additionally tests all 56 never-trained triples. **Three image suites** on dSprites and 3D Shapes. Encode a real image, apply $T(\Delta)$ to the learned latent for a factor change Δ, decode, and score pixel error against the true transformed image; OOD means never-trained *types* of two-factor change. Two of the three add a categorical shape factor. **Content.** A small diffusion transformer generates dSprites images from a full factor specification, so the condition names what to draw rather than how to change it. **Rollouts.** A world model applies the conditioning arm recurrently, one action per step, tested at horizons 10 and 20 after training on 3. **The successor.** Complex FiLM, through the dSprites and content setups. **Guidance.** Condition-dropout training, then classifier-free guidance at strength w for FiLM against condition powering for Complex FiLM, scored against deterministic ground truth at each strength.

5 Results

(1) When conditions form a group, composition is exact to numerical precision. On the hidden-basis suite, CGA reaches 3.3×10^{-8} error on unseen pairs and 5.3×10^{-8} on never-trained triples; the best baseline sits five orders of magnitude higher and every baseline degrades as combinations lengthen. Learned angles match the true generators to 2×10^{-4} rad: reading the weights tells you which plane each condition rotates, and how fast (Figure 4).

(2) The advantage survives learned representations. On dSprites every arm fits the training distribution equally well, which isolates composition. CGA reduces error on unseen combinations by 53.8% against the strongest baseline, at 39× fewer conditioning FLOPs, and by 42.9% once a categorical shape factor is added. The same result in a second form: going from trained to unseen combinations multiplies CGA's error by 1.25×, or 1.63× with the shape factor, and every baseline's by 2.1–3.3×. We call that multiplier the *compositional gap*.

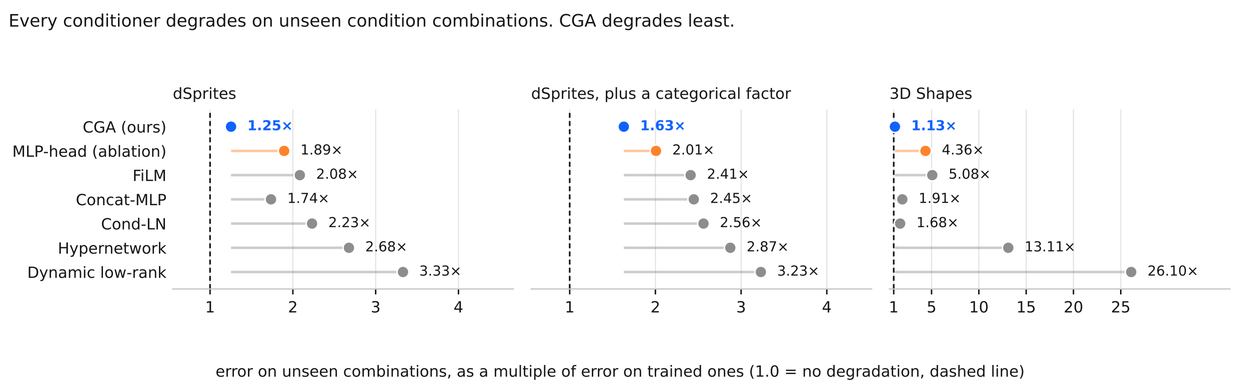

Figure 1: How much worse each conditioner gets when conditions are combined in ways it never saw, as a multiple of its own error on trained combinations. A value of 1.0, the dashed line, would mean no degradation at all. CGA sits closest to it on all three image suites, and the two most expressive conditioners sit furthest: their capacity goes into memorising the training combinations rather than generalising past them.

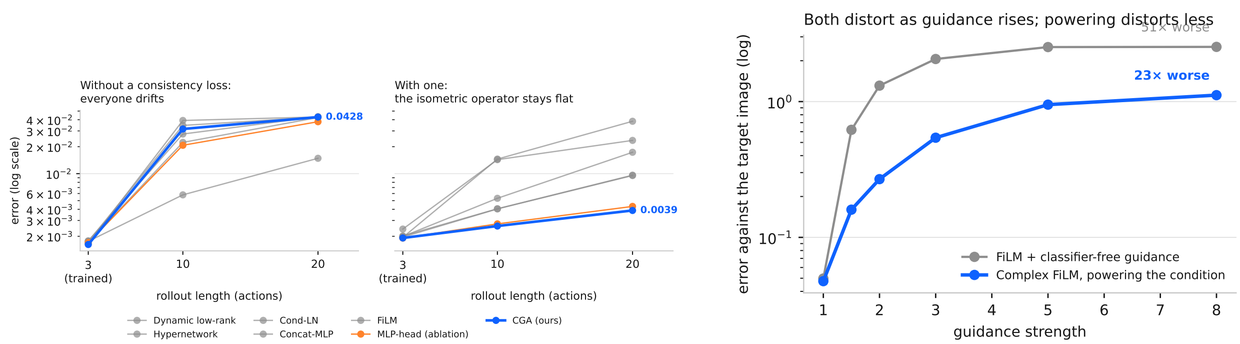

Figure 2: **Left:** error against rollout length, trained only to length 3. Without a consistency loss every conditioner drifts at much the same rate. Adding one separates them: the isometric operator flattens out while the baselines keep climbing. **Right:** what turning guidance up does to the generated image. Both distort, since the target here is deterministic and there is no distribution to sharpen, but classifier-free guidance distorts twice as fast.

(3) The linear map is the mechanism. The ablation keeps the basis and the operator family, spends more compute, and swaps only the linear map for an MLP. It loses by $4.3{\times}10^4\times$ on synthetic triples and 1.55× on dSprites.

(4) More capacity makes it worse. The two hypernetwork arms can express any operator. They post the largest compositional gaps of any arm on every image suite: 2.7–3.3× on dSprites and 13–26× on 3D Shapes, against CGA's 1.13–1.63×. Their capacity memorizes the training combinations.

(5) Three negatives, scored by the criteria we registered in advance. On 3D Shapes, CGA has the lowest error on unseen combinations (87.5% below the hypernetwork) but its training error is 1.46× the hypernetwork's, which trips the pre-registered no-regression criterion. In the content suite (the content suite), rotation conditioning fails: pixel MSE 0.31 against FiLM's 0.04, an 8× regression and the ablation fails identically: a rotation moves information and cannot add any, so it cannot carry "generate this image." In the rollout suite, every arm compounds error at the same rate; each step adds a small mismatch between the predicted and the true latent. A contracting operator shrinks past mismatches; an isometry preserves them, so they accumulate.

(5a) It transfers to physics, and the fit criterion is what stops it. All the evidence above comes from benchmarks built for studying disentanglement. On a 2D Navier-Stokes surrogate conditioned on physical parameters — viscosity, drag, forcing, background advection, with the initial vorticity field supplying the content and the parameters only modulating the evolution — CGA's error on unseen parameter pairs is 57.5% below the best unstructured arm at $\delta = -1.00$: every seed beat every seed, using 7× fewer conditioning parameters and 1.11× FiLM's cost. It is nonetheless a KILL, on in-distribution fit at 1.118× against a 1.10× ceiling.

The fit penalty shrinks as the task moves away from asking the operator to supply content (1.46× on 3D Shapes, 2.43× on a frozen latent, 1.118× here), but it does not vanish. Two relaxations that keep exact composition were pre-registered against it: giving the algebra a scaling generator, and conjugating the orthogonal basis by a learned diagonal so the rotation acts in a stretched metric. On validation the conjugated operator appeared to remove the penalty entirely (0.962×); on the held-out split over ten seeds it reached 1.1007× and failed the same criterion by 0.065 percentage points. Relaxing the *metric* is not the fix.

(5a′) Complex FiLM removes the fit penalty, and passes the same gate. Gated on fresh seeds that had never been trained, Complex FiLM passes every criterion on the physics task: 45.4% below FiLM and 64.6% below the best unstructured arm at $\delta = -1.00$, with an in-distribution fit ratio of **0.938**× — the first arm in this program to fit *better* than the unstructured baseline rather than merely close to it — at 0.84× FiLM's cost. The margin replicates across independent seed sets (+65.3% and 1.030× on the first, +64.6% and 0.938× on the second), which matters because the conjugated operator looked like a fix on validation and was not.

So the penalty was the isometry, and the remedy is a magnitude channel rather than a stretched metric. This is a claim about Complex FiLM and not about the pure group operator: its magnitude head is nonlinear in c, so composition is exact on the phase channel only. It buys in-distribution fit by spending half the exactness guarantee, and the arm that keeps the guarantee on both channels is the one that failed this criterion twice.

(5b) The advantage requires a co-trained representation. The sharpest bound comes from a controlled pair. The same dataset, the same held-out attribute pairs, the same arms and the same budget counter were run twice, differing in one variable: whether the representation is trained alongside the operator. Co-trained, CGA's error on unseen combinations is 87.5% below the hypernetwork's. Frozen — a plain autoencoder trained once for reconstruction, then shared unchanged by every arm, with scoring in its latent space so the decoder cannot flatter anyone — CGA is 60.1% *worse* than that hypernetwork, while using 48× fewer conditioning parameters, and fits 2.43× worse in distribution against a 1.10× ceiling. Complex FiLM fails there too, so this is not the rotation channel alone.

The reading is that the operator's advantage is not a property of the operator. It requires the representation to be shaped by that operator during training. Bolted onto a frozen general-purpose encoder — the pattern latent-space editing actually deploys — it loses. CGA still has the smallest compositional gap of any arm that fits (1.39× against the hypernetwork's 2.11×): it degrades most gracefully and still loses, because grace does not recover a fit handicap.

(6) A consistency loss completes the rollout recipe. Training with $\lambda \| T(a) z_s - z_{s+1} \|^2$, applied to every arm equally, inverts the picture: the rotation operator's horizon-20 error goes flat at 0.0039 (growth 2.0×) against the hypernetwork's 0.038 (growth 19.8×) and FiLM's 0.017 (Figure 2, left). The loss is standard in model-based RL [19], and rotation transitions with latent-prediction objectives appear in the homomorphism autoencoder [18]; our contribution is to separate the two effects: adding the same loss to every arm at the same budget shows that neither the loss nor the isometric operator fixes rollouts alone, and that they work only together. A null result sharpens this. We tried the obvious alternative and it failed: shrinking the latent by a fixed factor at every step, swept from 0.003 to 0.1, leaves horizon-20 error unchanged. Whatever removes the accumulated mismatch has to depend on the state rather than be a constant. A separate single-step criterion fails, so the rollout suite counts as a negative in Table 1 even though its horizon-20 result is favorable.

(7) The negatives specify the fix, and it works. Complex FiLM multiplies feature pairs by $m\, e^{i\theta}$. Under two pre-registered gates it beats FiLM by 36% on unseen change types at fit parity and 0.84× FiLM's cost, and matches FiLM on content generation at 0.97× its error (pre-registered non-inferiority margin 1.10×), where the pure rotation failed by 7×. Giving the magnitude channel a linear map instead of an MLP fails on that same content metric (0.19 against FiLM's 0.04): content needs the expressive head. FiLM is the magnitude channel on its own; the full complex multiplication does both jobs.

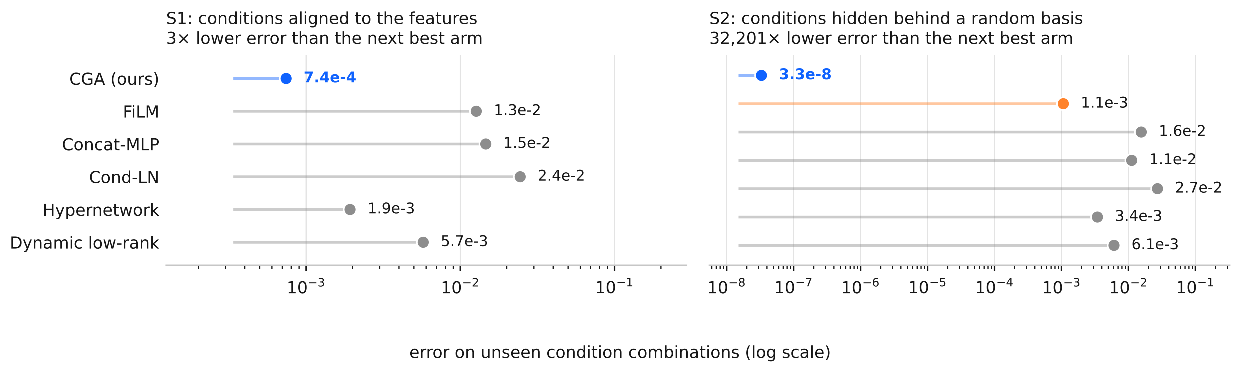

Figure 3: On the synthetic suites, where the conditions form a group exactly, the gap is not a matter of percentages. Each dot is one conditioner's error on unseen combinations; note the log axis. Hiding the conditions behind a random basis (right) costs CGA nothing, because it learns the basis, while the gap to the next best arm grows to five orders of magnitude.

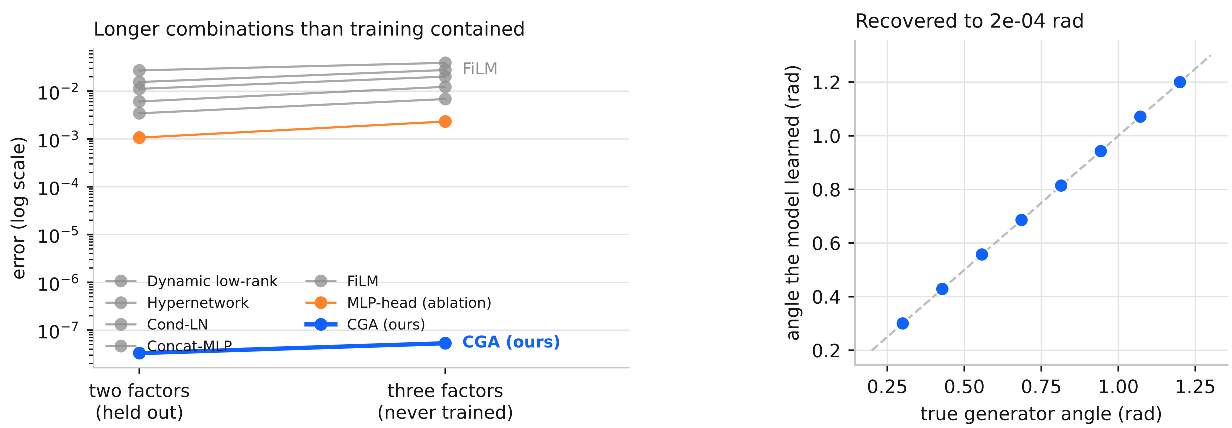

Figure 4: **Left:** what happens when a test combination is longer than anything in training. Every baseline gets worse; CGA does not move, because composition is a property of its parametrization rather than something it had to learn. **Right:** the angles the model learned for each condition, against the true generators of the data. Reading the weights tells you which plane a condition rotates, and how fast.

(8) Guidance without the second pass. CFG's distortion at high scales is established [24]; our benchmark isolates it against an exact alternative, because the conditioning target is deterministic. CFG's error grows 50.7× from strength 1 to 8; powering the condition through the phase channel grows 23.7× (p=1.6×10^{-4}), stays 2–5× lower at every strength above 1, matches at native strength (0.96×), and needs no unconditional pass and no distillation [25] (Figure 2, right). Neither method improves over strength 1 here: the target is deterministic, so there is no distribution to sharpen, exactly as the suite's registered caveat stated.

6 Practical guidance

Three rules for practitioners, from the eight suites. **Use phase conditioning when your conditions are changes that combine.** Edit deltas, attribute offsets, pose or camera adjustments, applied once. Cost matches FiLM, fit matches, and error on never-trained combinations is 1.8–8× lower on real images (42.9–87.5% reduction against the strongest baseline) and up to 10^5× lower when the conditions form a group exactly. FiLM gives no warning when it fails this way: its training loss looks normal, and only the unseen combinations are wrong. CGA's composition law rules that failure out by construction. **For conditions that say what to generate, use Complex FiLM or keep FiLM.** Pure rotation fails there (7×); Complex FiLM matches FiLM on content (0.97×) while keeping the composition wins, so it is the candidate drop-in at our evidence scale. **In rollouts, pair an isometric operator with a consistency loss.** Without the loss, all conditioners compound error at the same rate; with it, the rotation operator stays flat where FiLM grows 4.5× and a hypernetwork 10×.

7 Limitations

All evidence comes from controlled 64×64 benchmarks with known factors, width-128 models, and a single conditioning site; nothing here is validated at production scale, and the natural deployment experiment is the adaLN site of a text-conditioned diffusion transformer. Learned latents do not support exact group action, so the advantage on real images is the 1.8–8× error reduction quoted above, not the 10^5× of the synthetic suites. 3D Shapes shows orthogonality can cost in-distribution fit on rich scenes. Conditions far from any group action, such as open-ended text, remain untested; the learned-condition-geometry extension is future work.

The guidance result does not transfer to a multimodal target. Our guidance benchmark measures distortion against a deterministic target, so it cannot see the distribution-sharpening CFG exists to provide. We built a second harness to check: condition on shape, scale and orientation and leave position free, making $p(x \mid c)$ uniform over 64 cells exactly, so fidelity and diversity are separately measurable against a target known in closed form. An unregistered single-seed pilot on it — no gate, no verdict claimed — points against the method on all three axes we could measure. At matched fidelity CFG retains *more* diversity (coverage 0.946 at fidelity 0.919, against ours 0.920 at a better 0.828), so CFG dominates the frontier. The strength knob is not monotone: fidelity runs 0.139, 1.979, 1.300, 0.828, 0.733, 0.800 as α goes 1 to 8, which is the rotation channel wrapping, since $T(c)^\alpha$ turns by $\alpha\theta$ and turning further is not the same as conditioning harder. And the base model fits 33% worse before any guidance is applied. We therefore claim the guidance result only as stated — less distortion at equal cost on a deterministic target — and not as a general replacement for CFG. Establishing that would need the frontier comparison to come out the other way, and in our pilot it does not.

Reproducibility statement. We release the code, the pre-registrations, and the raw logs. Every suite has a specification committed before its decision run; a registry maps each suite label in this paper to the module that produced it, the log it wrote, and its verdict, so a replicator reproduces any row with a single command. Every number, table, and figure here regenerates from those append-only logs. Unit tests pin the operator invariants (identity at initialization and at $c = 0$, orthogonality, exact composition), the budget counters, and the registry itself. All amendments and one accounting erratum, a FLOP undercount our own audit caught that downgraded a favorable verdict, are disclosed with the results.

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